

1. Explain and demonstrate the use of **1D Arrays, Multidimensional Arrays, and Irregular Arrays** in Java with suitable examples.
2. Compare and contrast **ArrayList** and **Vector** classes in Java.
3. Explain the following operations of the **String** class with examples:
 - Creation of strings using constructors and literals
 - Concatenation and comparison
 - Case conversion and searching
 - Splitting and joining strings
4. Discuss the importance of the **StringBuffer** class in Java.
5. Write a Java program that demonstrates:
 - Multilevel inheritance (e.g., Person → Employee → Manager).
 - Constructor calling sequence using super().
 - Method overriding in subclasses.
 - The effect of declaring a method as final.

1. Briefly discuss about Interfaces and extending multiple interfaces.
2. Write a Java program to demonstrate **multiple inheritance using interfaces Printable and Showable**. Add a **nested interface** inside Printable and implement it in a class. Show how nested interface methods are accessed.
3. Create interfaces **Vehicle** and **ElectricVehicle** (extends Vehicle). Implement them in TATACar class and show how **interface references** invoke methods of base and extended interfaces.
4. Demonstrate the Packages and types of Packages with example.
5. What is an Exception Handling? Discuss about the Five Key words of Exception Handling.
6. Write a Java program to demonstrate **multiple catch blocks, throw, throws, and finally**.
7. Briefly discuss about Concurrency(Multithreading) with example and types of creating a Thread.
8. Write short note on Inner Classes and Lambda Expression in java programing